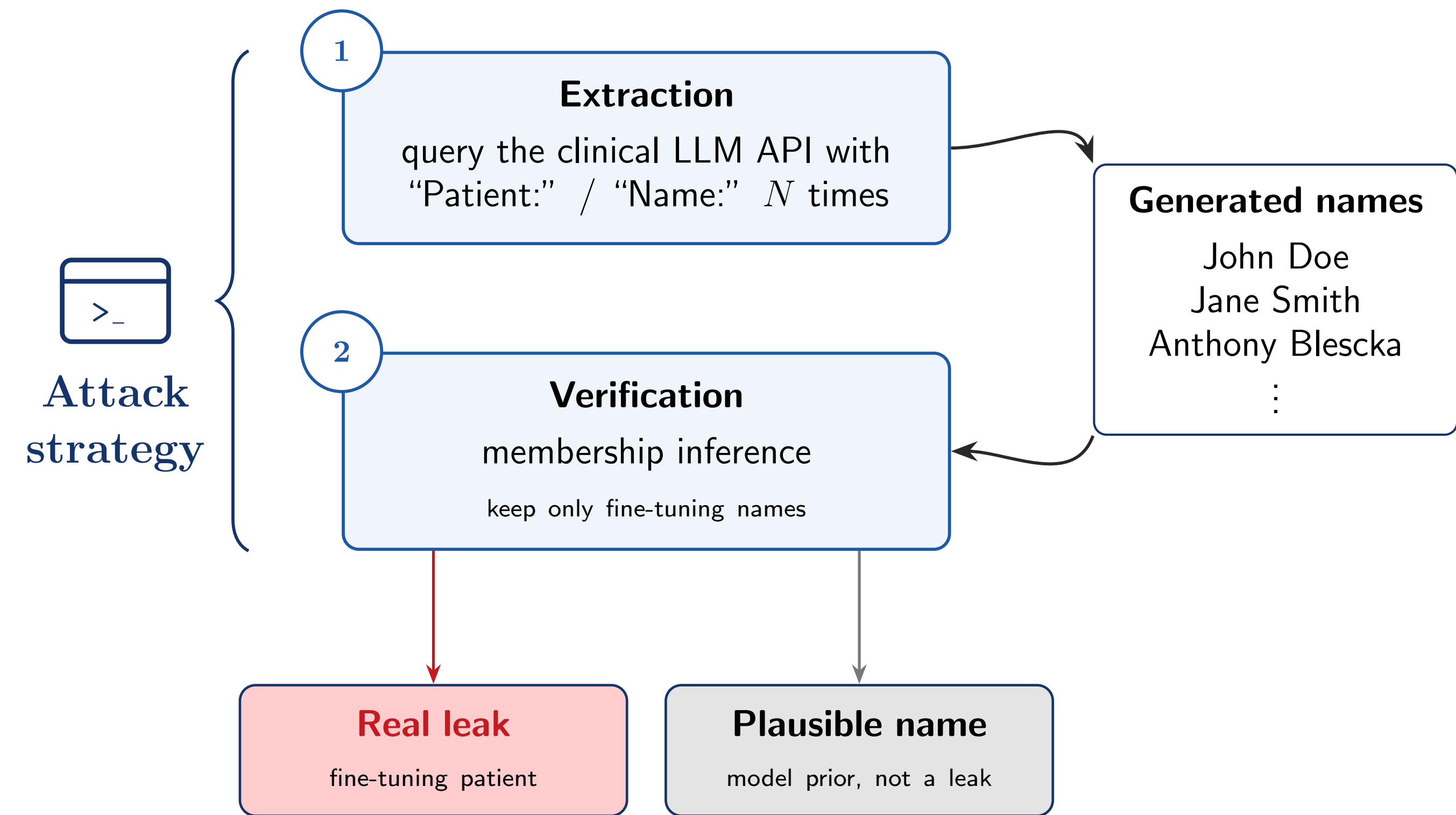




1 Motivation

- Clinical notes are rich data for LLM fine-tuning, but they contain sensitive patient information.
- De-identification is never perfect: state-of-the-art automated tools reach only $\sim 99\%$ F1-score (Tannier et al., 2024), so **residual direct identifiers survive** in the fine-tuning text.
- LLMs **memorize and leak** training data across prompts and decoding strategies (Thakkar et al., 2023; Pan et al., 2024), so a fine-tuned clinical model can emit *real* patient names.
- Yet prior studies underestimate leakage because they don't account for the attacker's query budget (Lehman et al., 2021; Huang et al., 2022).



How can we realistically measure *how much* an attacker can recover of the residual direct identifiers (DI) in a fine-tuned clinical LLM given a limited query budget?

2 Setup

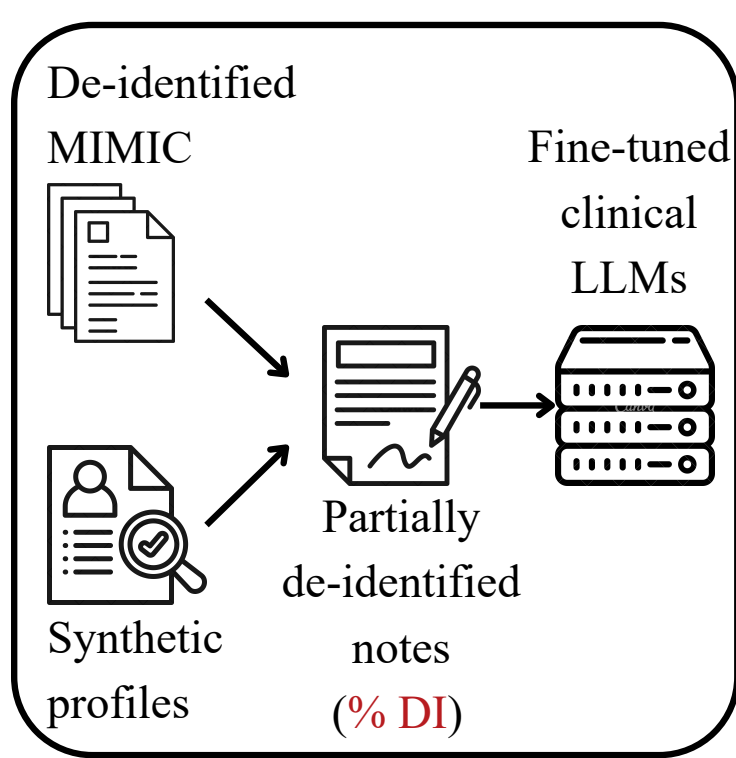
Example of a note with residual direct identifiers

Name: _____ MRN: _____

Allergies: None

Chief Complaint: Chest Pain

The patient **John Doe** has a history of hypertension and presents with acute onset...



- Dataset: $\sim 300k$ MIMIC-IV notes (Johnson et al., 2023).
- Fine-tuning: full Supervised Fine-Tuning (SFT) of Llama-3 (Llama Team, AI Meta, 2024) for clinical note generation.

DI rate	# Distinct names
1%	9,710
5%	40,467
10%	68,730
100%	234,904

- DI rate: the share of original name mentions that de-identification leaves behind ($10\% \Rightarrow 1$ in 10 names survives).

3 Methods

Threat model: We assume

- Gray-box** access via a provider API: token probabilities, but not weights.
- Access to a small labeled set of member / non-member names to train the verifier.

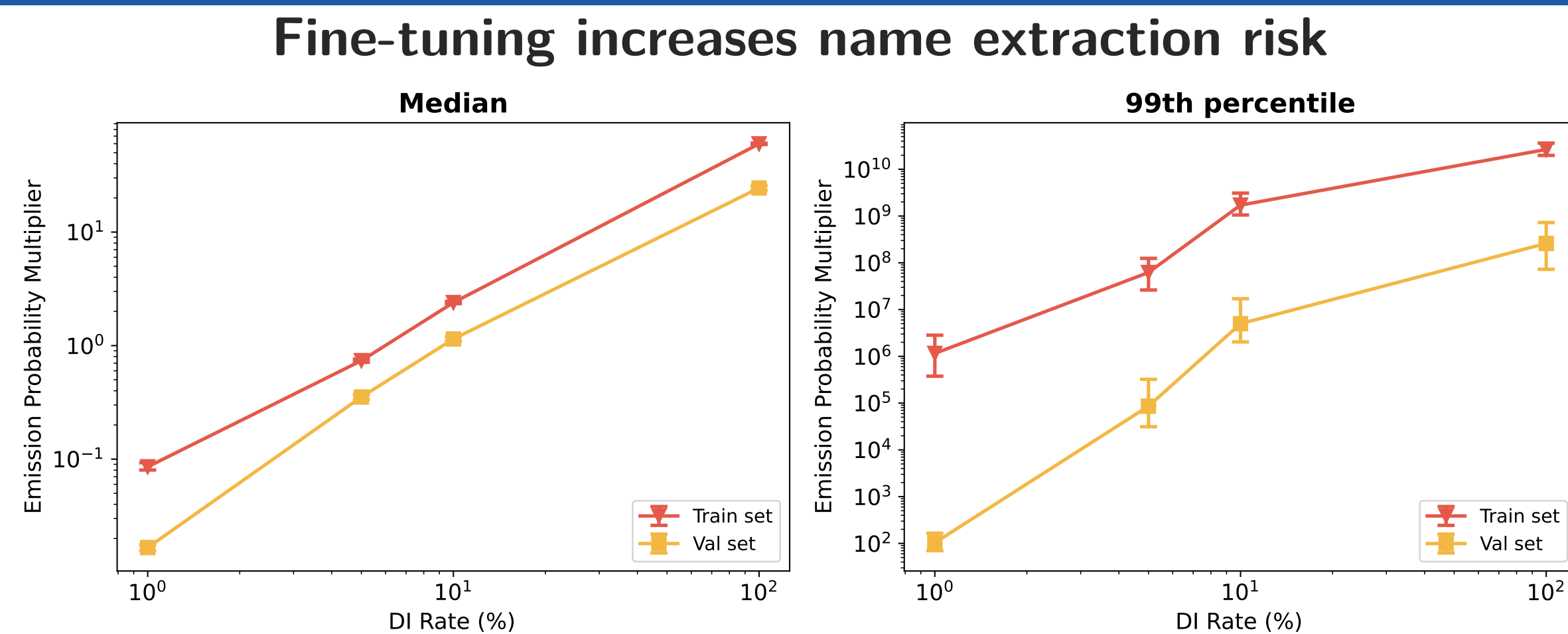
Verified Extraction:

- Extraction: we repeatedly query the model (Name:) to surface candidate names.
- Verification: a Membership Inference Attack (MIA) classifier keeps **only names attributable to the fine-tuning data**, discarding the LLM prior.

Risk evaluation:

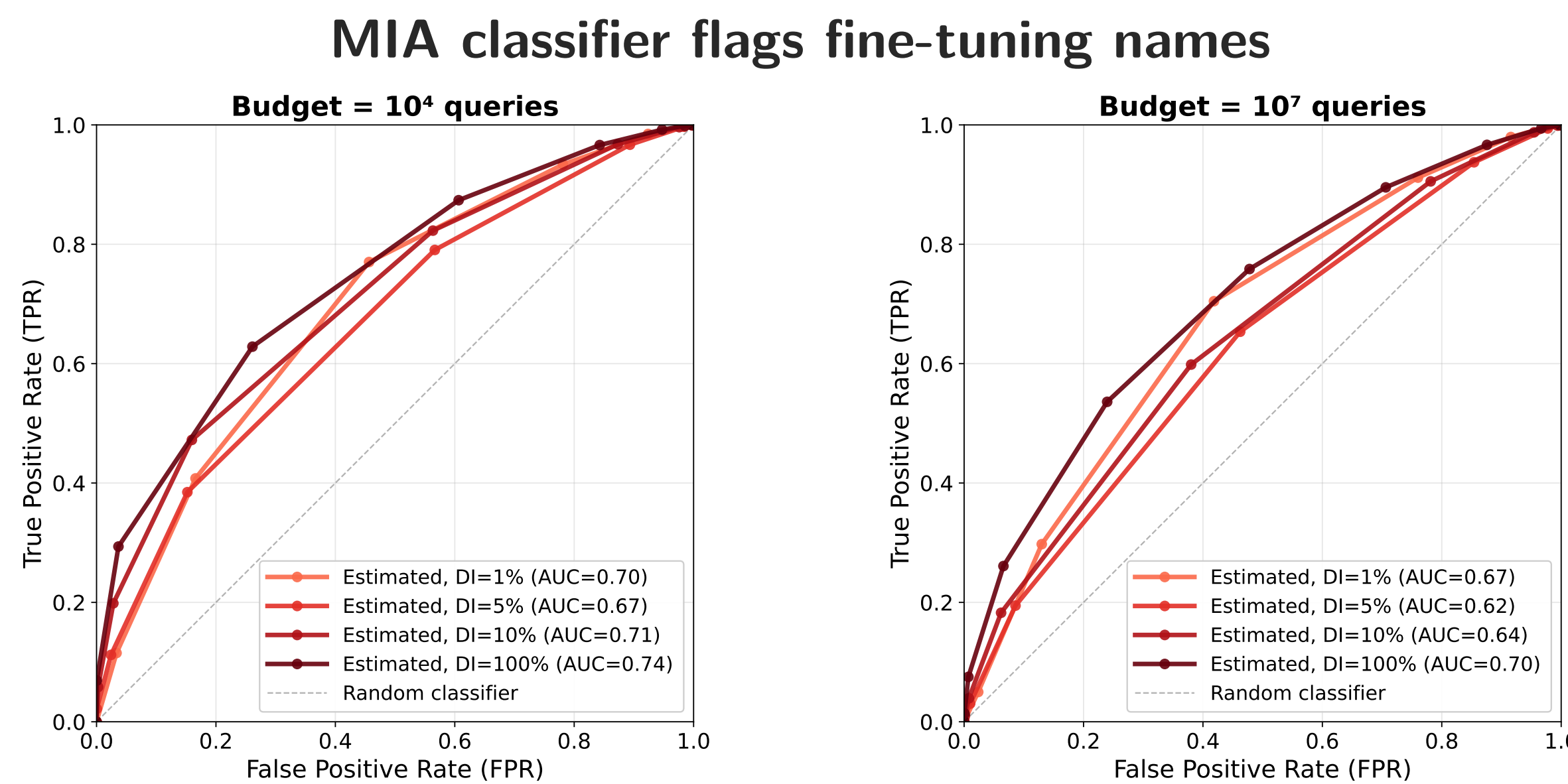
- Analytical: per-name emission probabilities (teacher-forcing) + MIA scores give **budget-conditioned leakage curves** in closed form, no sampling needed.
- Empirical: we query the model N times, run the generated names through the verifier, and count the verified recoveries.

4 Results



Emission Probability Multiplier per name (how much more likely the extraction for the fine-tuned vs the base model for each name is). 1B model.

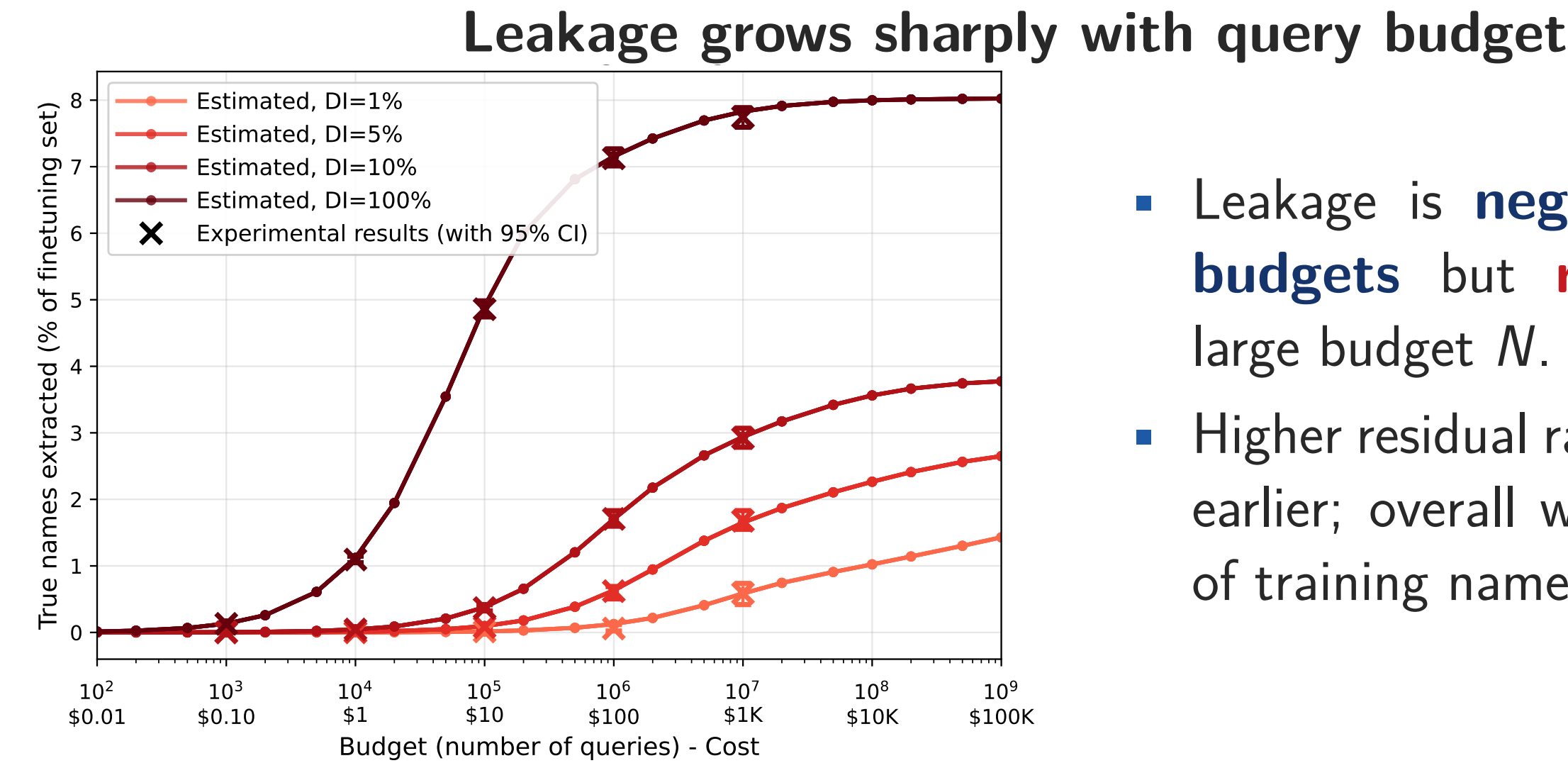
- Median: the gap between the train and val multipliers is small, and the order of magnitude is close to 1: the model updates its name prior.
- Memorization appears in the **train-val gap in the tail** of the emission: the risk concentrates in a few strongly amplified names.



Budget-conditioned ROC of the MIA verifier at $N=10^4$ and $N=10^7$, across residual DI rates.

- Among the generated names, membership inference separates true fine-tuning names from names originating from the LLM prior.
- As N grows, the stream adds lower-probability names that are harder to classify, so AUC drops: the privacy-risk is **budget-conditioned**.

4 Results (continued)



1B model. Curves: analytical. Markers: empirical evaluations by generating N completions.

Scaling & attack robustness

DI rate	N	1B (%)	8B (%)	Ratio
1%	10^5	0.01	2.24	152.0×
1%	10^6	0.06	6.83	116.0×
5%	10^5	0.08	1.30	16.0×
5%	10^6	0.62	5.51	8.9×
10%	10^5	0.37	2.85	7.8×
10%	10^6	1.71	9.80	5.7×
100%	10^5	4.86	15.40	3.2×
100%	10^6	7.13	24.26	3.4×

Verified names recovered (% of fine-tuning set).
Ratio = 8B / 1B.

Scaling:

- The 8B model leaks more than the 1B model.
- Increasing model size raises leakage the most at low DI rates.

Prompt sensitivity:

- We compared Name: and Patient:.
- The second prompt Patient: recovers nearly as many names as Name: ($\leq 4\%$ difference at $N=10^{10}$).

Verifier ablation:

- Using additional features for the verifier like base-model name emission probabilities adds no signal beyond the fine-tuned-model probabilities of the names.
- Need at least $\geq 1\%$ of labeled candidates for proper FPR calibration.

5 Generalizable Insights about MLHC and key Takeaways

- Verified Extraction** realistically audits clinical LLMs by combining repeated query extraction with membership verification.
- Even with modest query budgets, an attacker can recover a concerning number of direct identifiers. For instance, at a 10% residual rate, for Llama3-1B, an attacker recovers about 2000 names with 95% precision using $\sim 10^7$ queries, $\sim \$1k$ at current API prices.
- Although de-identification results in a low direct identifier rate, measurable leakage persists and increases most sharply with scaling in the low-DI regime.

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